

1.1 AI Pervasiveness

Artificial Intelligence (AI) has emerged as one of the most transformative forces in human history. From the moment we wake up to when we go to sleep, AI systems invisibly shape our world. They manage traffic flows in our cities, optimize power distribution across electrical grids, and enable billions of wireless devices to communicate seamlessly. In hospitals, AI analyzes medical images and helps doctors diagnose diseases. In research laboratories, it accelerates scientific discovery by simulating molecular interactions and processing vast datasets from particle accelerators. In space exploration, it helps rovers navigate distant planets and telescopes detect new celestial phenomena.

Throughout history, certain technologies have fundamentally transformed human civilization, defining their eras. The 18th and 19th centuries were shaped by the Industrial Revolution, where steam power and mechanization transformed how humans could harness physical energy. The 20th century was defined by the Digital Revolution, where the computer and internet transformed how we process and share information. Now, the 21st century appears to be the era of Artificial Intelligence, a shift noted by leading thinkers in technological evolution (Byzantion and McAfee 2014; Domingos 2016).

The vision driving AI development extends far beyond the practical applications we see today. We aspire to create systems that can work alongside humanity, enhancing our problem-solving capabilities and accelerating scientific progress. Imagine AI systems that could help us understand consciousness, decode the complexities of biological systems, or unravel the mysteries of dark matter. Consider the potential of AI to help address global challenges like climate change, disease, or sustainable energy production. This is not just about automation or efficiency—it's about expanding the boundaries of human knowledge and capability.

The impact of this revolution operates at multiple scales, each with profound implications. At the individual level, AI personalizes our experiences and augments our daily decision-making capabilities. At the organizational level, it transforms how businesses operate and how research institutions make discoveries. At the societal level, it reshapes everything from transportation systems to healthcare delivery. At the global level, it offers new approaches to addressing humanity's greatest challenges, from climate change to drug discovery.

What makes this transformation unique is its unprecedented pace. While the Industrial Revolution unfolded over centuries and the Digital Revolution over decades, AI capabilities are advancing at an extraordinary rate. Technologies that seemed impossible just years ago, including systems that can understand human speech, generate novel ideas, or make complex decisions, are now commonplace. This acceleration suggests we are only beginning to understand how profoundly AI will reshape our world.

We stand at a historic inflection point. Just as the Industrial Revolution required us to master mechanical engineering to harness the power of steam and machinery, and the Digital Revolution demanded expertise in electrical and computer engineering to build the internet age, the AI Revolution presents us with a new engineering challenge. We must learn to build systems that

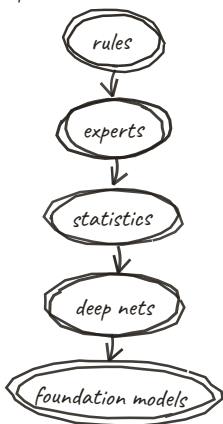
And drives the cars too - Waymo passed 1M+ fully driverless rides per month by 2025. ☆

No longer 'appears' - settled. ChatGPT hit 100M users in 2 months (2023), fastest consumer app ever.

Citing 2014 for the AI century? The 2020s literature IS the story now.

Already begun: AlphaFold cracked protein folding - 2024 Nobel Prize in Chemistry.

the whole plot, one line:



can learn, reason, and potentially achieve superhuman capabilities in specific domains.

1.2 AI and ML Basics

The exploration of artificial intelligence's transformative impact across society presents a fundamental question: How can we create these intelligent capabilities? Understanding the relationship between AI and ML provides the theoretical and practical framework necessary to address this question.

Artificial Intelligence represents the systematic pursuit of understanding and replicating intelligent behavior—specifically, the capacity to learn, reason, and adapt to new situations. It encompasses fundamental questions about the nature of intelligence, knowledge, and learning. How do we recognize patterns? How do we learn from experience? How do we adapt our behavior based on new information? AI as a field explores these questions, drawing insights from cognitive science, psychology, neuroscience, and computer science.

Machine Learning, in contrast, constitutes the methodological approach to creating systems that demonstrate intelligent behavior. Instead of implementing intelligence through predetermined rules, machine learning systems utilize gradient descent¹ and other optimization techniques to identify patterns and relationships. This methodology reflects fundamental learning processes observed in biological systems. For instance, object recognition in machine learning systems parallels human visual learning processes, requiring exposure to numerous examples to develop robust recognition capabilities. Similarly, natural language processing systems acquire linguistic capabilities through extensive analysis of textual data.

The relationship between AI and ML exemplifies the connection between theoretical understanding and practical engineering implementation observed in other scientific fields. For instance, physics provides the theoretical foundation for mechanical engineering's practical applications in structural design and machinery, while AI's theoretical frameworks inform machine learning's practical development of intelligent systems. Similarly, electrical engineering's transformation of electromagnetic theory into functional power systems parallels machine learning's implementation of intelligence theories into operational ML systems.

💡 AI and ML: Key Definitions

- **Artificial Intelligence (AI):** The goal of creating machines that can match or exceed human intelligence—representing humanity's quest to build systems that can think, reason, and adapt.
- **Machine Learning (ML):** The scientific discipline of understanding how systems can learn and improve from experience—providing the theoretical foundation for building intelligent systems.

The emergence of machine learning as a viable scientific discipline approach to artificial intelligence resulted from extensive research and fundamental

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➔

Really
AdamW since
2017. Plain
GD is
pedagogy.

'Numerous' =
15
TRILLION
tokens now
(Llama 3).

¹ Gradient Descent: An optimization algorithm that iteratively adjusts model parameters to minimize prediction errors by following the gradient (slope) of the error surface, similar to finding the bottom of a valley by always walking downhill.

² | Deep Reinforcement Learning: A machine learning approach that combines deep neural networks with reinforcement learning principles, allowing agents to learn optimal actions through trial and error interaction with an environment while receiving rewards or penalties.

paradigm shifts in the field. The progression of artificial intelligence encompasses both theoretical advances in understanding intelligence and practical developments in implementation methodologies. This development mirrors the evolution of other scientific and engineering disciplines—from mechanical engineering’s advancement from basic force principles to contemporary robotics, to electrical engineering’s progression from fundamental electromagnetic theory to modern power and communication networks. Analysis of this historical trajectory reveals both the technological innovations leading to current machine learning approaches and the emergence of deep reinforcement learning that inform contemporary AI system development.

Now it's self-supervised transformers; RL lives on in post-training (RLHF).



Self-Check: Question 1.1

1. Explain how the relationship between AI and ML is similar to the relationship between physics and mechanical engineering.
2. True or False: Machine Learning systems implement intelligence through predetermined rules.

See Answer →

1.3 AI Evolution

³ | Perceptron: The first artificial neural network—a simple model that could learn to classify visual patterns, similar to a single neuron making a yes/no decision based on its inputs.

The evolution of AI, depicted in the timeline shown in Figure 1.1, highlights key milestones such as the development of the perceptron³ in 1957 by Frank Rosenblatt, a foundational element for modern neural networks. Imagine walking into a computer lab in 1965. You’d find room-sized mainframes running programs that could prove basic mathematical theorems or play simple games like tic-tac-toe. These early artificial intelligence systems, while groundbreaking for their time, were a far cry from today’s machine learning systems that can detect cancer in medical images or understand human speech. The timeline shows the progression from early innovations like the ELIZA chatbot in 1966, to significant breakthroughs such as IBM’s Deep Blue defeating chess champion Garry Kasparov in 1997. More recent advancements include the introduction of OpenAI’s GPT-3 in 2020 and GPT-4 in 2023, demonstrating the dramatic evolution and increasing complexity of AI systems over the decades.

67 yrs on: TWO Nobels for neural nets in 2024.

2025: IMO gold. Tic-tac-toe to Olympiad in 60 yrs.

Let’s explore how we got here.

1.3.1 Symbolic AI Era

The story of machine learning begins at the historic Dartmouth Conference in 1956, where pioneers like John McCarthy, Marvin Minsky, and Claude Shannon first coined the term “artificial intelligence.” Their approach was based on a compelling idea: intelligence could be reduced to symbol manipulation. Consider Daniel Bobrow’s STUDENT system from 1964, one of the first AI programs that could solve algebra word problems. It was one of the first AI programs to demonstrate natural language understanding by converting English text into algebraic equations, marking an important milestone in symbolic AI.

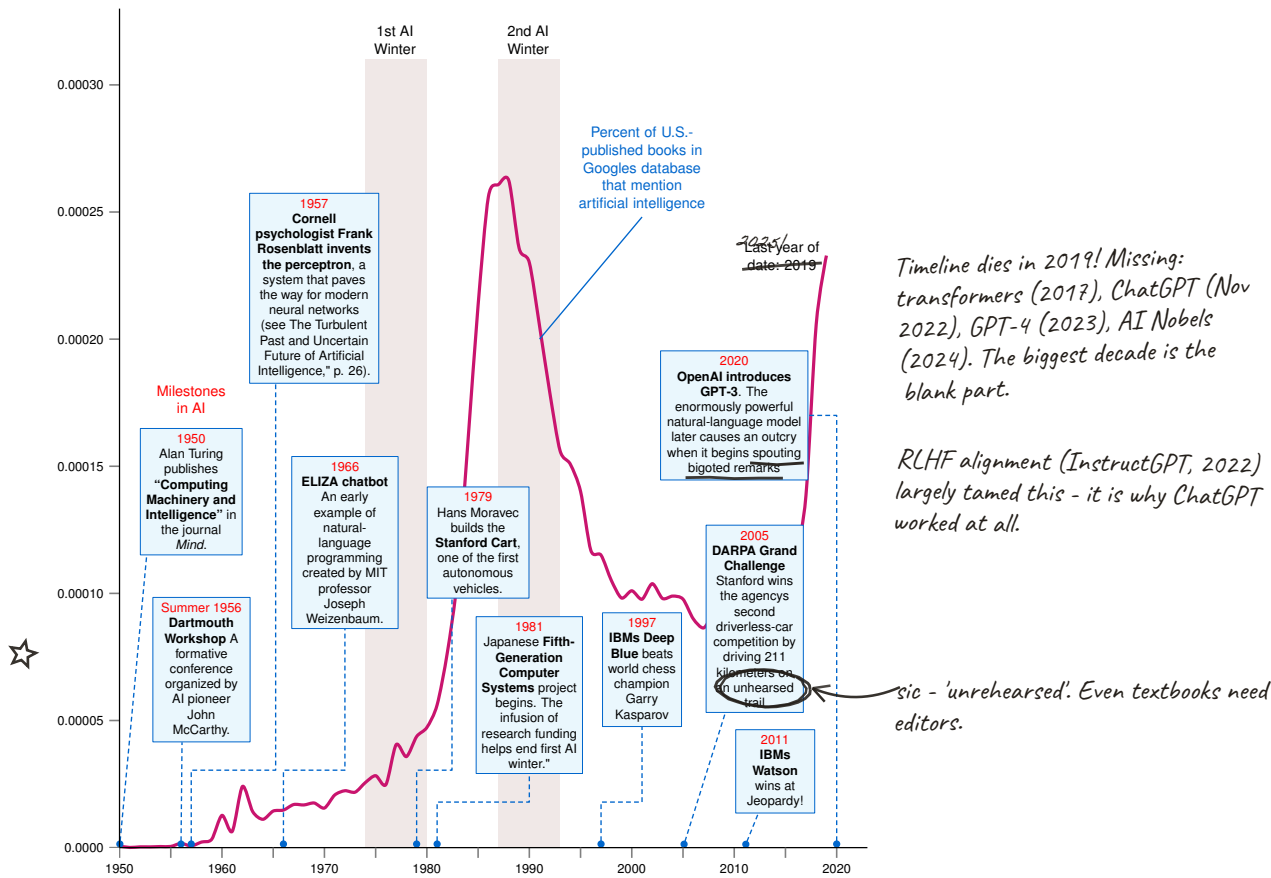


Figure 1.1: AI Development Timeline: Early AI research focused on symbolic reasoning and rule-based systems, while modern AI leverages data-driven approaches like neural networks to achieve increasingly complex tasks. This progression exposes a shift from hand-coded intelligence to learned intelligence, marked by milestones such as the perceptron, deep blue, and large language models like GPT-3.

i Example: STUDENT (1964)

Problem: "If the number of customers Tom gets is twice the square of 20% of the number of advertisements he runs, and the number of advertisements is 45, what is the number of customers Tom gets?"

STUDENT would:

1. Parse the English text

2. Convert it to algebraic equations
3. Solve the equation: $n = 2(0.2 \times 45)^2$
4. Provide the answer: 162 customers

Early AI like STUDENT suffered from a fundamental limitation: they could only handle inputs that exactly matched their pre-programmed patterns and rules. Imagine a language translator that only works when sentences follow perfect grammatical structure; even slight variations, such as changing word order, using synonyms, or natural speech patterns, would cause the STUDENT to fail. This “brittleness” meant that while these solutions could appear intelligent when handling very specific cases they were designed for, they would break down completely when faced with even minor variations or real-world complexity. This limitation wasn’t just a technical inconvenience—it revealed a deeper problem with rule-based approaches to AI: they couldn’t genuinely understand or generalize from their programming, they could only match and manipulate patterns exactly as specified.

The exact critique aimed at LLMs too - 'stochastic parrots' (2021). Reasoning models weakened it.

1.3.2 Expert Systems Era

By the mid-1970s, researchers realized that general AI was too ambitious. Instead, they focused on capturing human expert knowledge in specific domains. MYCIN, developed at Stanford, was one of the first large-scale expert systems designed to diagnose blood infections.

i Example: MYCIN (1976)

Rule Example from MYCIN:

IF

The infection is primary-bacteremia

The site of the culture is one of the sterile sites

The suspected portal of entry is the gastrointestinal tract

THEN

Found ~~suggestive evidence (0.7) that infection is bacteroid~~

vs ZERO hand-written rules: GPT-4 passed US medical licensing exam questions (2023). Data beat rules.

Now inverted: the MODEL cannot explain itself. Interpretability is the new MYCIN problem (Anthropic circuits work, 2024-25).

While MYCIN represented a major advance in medical AI with its 600 expert rules for diagnosing blood infections, it revealed fundamental challenges that still plague ML today. Getting domain knowledge from human experts and converting it into precise rules proved incredibly time-consuming and difficult—doctors often couldn’t explain exactly how they made decisions. MYCIN struggled with uncertain or incomplete information, unlike human doctors who could make educated guesses. Perhaps most importantly, maintaining and updating the rule base became exponentially more complex as MYCIN grew, as adding new rules frequently conflicted with existing ones, while medical knowledge itself continued to evolve. These same challenges of knowledge capture, uncertainty handling, and maintenance remain central concerns in modern machine learning, even though we now use different technical approaches to address them.